WORKED SOLUTION

STEP III 2004, Question 2

Pure • Asymptotes and Curve Sketching

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q2

The key idea

Write $f(x)$ as the line $y = x - 4$ plus a correction term. The correction is a perfect square on top, so it vanishes *twice* at one point, and that double zero is exactly what tangency means. For the roots of $f(x) = 0$, clear the denominator: you get a difference of two squares, which factorises at once.

Solution

Throughout, write

$$f(x) = (x-4) - \frac{16(2x+1)^2}{x^2(x-4)}.$$

Part (i)

The denominator vanishes at $x = 0$ and $x = 4$. The numerator $16(2x+1)^2$ is non-zero at both, so the vertical asymptotes are

$$x = 0 \qquad \text{and} \qquad x = 4.$$

For large $|x|$ the fraction has a quadratic on top and a cubic underneath:

$$\frac{16(2x+1)^2}{x^2(x-4)} = \frac{64x^2 + 64x + 16}{x^3 - 4x^2} \approx \frac{64}{x} \to 0 \quad \text{as } x \to \pm\infty,$$

so $f(x) - (x-4) \to 0$, and the oblique asymptote is

$$y = x - 4.$$

Ask yourself How can an asymptote also be a tangent?

An asymptote only describes what happens far away, so nothing stops the line from touching the curve at some finite point as well. To find where the line and curve meet, set $f(x) = x - 4$:

$$f(x) - (x-4) = -\frac{16(2x+1)^2}{x^2(x-4)} = 0 \iff x = -\tfrac{1}{2}.$$

Because the factor $(2x+1)$ is squared, $x = -\frac{1}{2}$ is a *double* root of $f(x) - (x-4) = 0$. This means the line touches the curve there. To check directly, let $D(x) = f(x) - (x-4) = (2x+1)^2 h(x)$ with $h(x) = -16/(x^2(x-4))$. The function h is differentiable near $x = -\frac{1}{2}$, so

$$D'(x) = 4(2x+1)h(x) + (2x+1)^2 h'(x),$$

which is 0 at $x = -\frac{1}{2}$. Hence $f'(-\frac{1}{2}) = 1$, the gradient of the asymptote. Also $f(-\frac{1}{2}) = -\frac{9}{2}$, which lies on $y = x - 4$. So the oblique asymptote is the tangent to the curve at $(-\frac{1}{2}, -\frac{9}{2})$.

Part (ii)

Put everything over the common denominator $x^2(x-4)$:

$$f(x) = \frac{x^2(x-4)^2 - 16(2x+1)^2}{x^2(x-4)} = \frac{[x(x-4)]^2 - [4(2x+1)]^2}{x^2(x-4)}.$$

The numerator is a difference of two squares:

$$\begin{aligned}[x^2 - 4x - 4(2x+1)][x^2 - 4x + 4(2x+1)] &= (x^2 - 12x - 4)(x^2 + 4x + 4) \\ &= (x^2 - 12x - 4)(x+2)^2.\end{aligned}$$

So

$$f(x) = \frac{(x+2)^2(x^2 - 12x - 4)}{x^2(x-4)}.$$

The factor $(x+2)^2$ shows that $x = -2$ is a double root of $f(x) = 0$, and $x = -2$ is not a zero of the denominator, as required. The other two roots come from $x^2 - 12x - 4 = 0$:

$$x = 6 \pm 2\sqrt{10} \approx 12.32,\ -0.32.$$

Part (iii)

We collect the features we need for the sketch.

Near $x = 0$. As $x \to 0$, $f(x) \approx -\dfrac{16}{x^2 \cdot (-4)} = \dfrac{4}{x^2}$, so $f(x) \to +\infty$ from *both* sides.

Near $x = 4$. As $x \to 4$, $f(x) \approx -\dfrac{16 \cdot 81}{16(x-4)} = -\dfrac{81}{x-4}$. So $f(x) \to +\infty$ as $x \to 4^-$ and $f(x) \to -\infty$ as $x \to 4^+$.

Which side of the oblique asymptote? $f(x) - (x-4) = -16(2x+1)^2/(x^2(x-4))$. This is positive for $x < 4$ (apart from $x = 0$ and $x = -\frac{1}{2}$) and negative for $x > 4$. So the curve

lies above $y = x - 4$ on the left of $x = 4$, touching it only at $x = -\frac{1}{2}$, and below it on the right.

Sign of f. From the factorised form, the factors $(x+2)^2$ and x^2 are never negative. So the sign of f is the sign of $(x^2 - 12x - 4)/(x - 4)$:

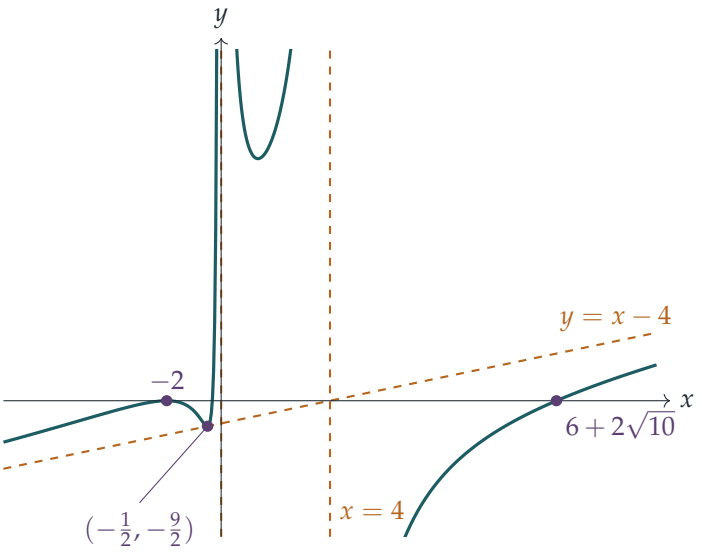
- $\mathrm{f} < 0$ for $x < 6 - 2\sqrt{10}$ (except $\mathrm{f}(-2) = 0$) and for $4 < x < 6 + 2\sqrt{10}$;
- $\mathrm{f} > 0$ for $6 - 2\sqrt{10} < x < 4$ ($x \neq 0$) and for $x > 6 + 2\sqrt{10}$.

The curve stays negative on both sides of $x = -2$ and touches the x-axis there, so $(-2, 0)$ is a local maximum.

Turning points. These aren't needed for a STEP sketch, but they make it more accurate. Differentiating gives

$$\mathrm{f}'(x) = 1 + \frac{16(2x+1)(2x^2+3x-8)}{x^3(x-4)^2}.$$

On $(-2,\, 6 - 2\sqrt{10})$ the curve starts and ends at height 0 and is negative in between, so it has a minimum there. Numerically this minimum is at about $(-0.51,\, -4.50)$, just left of the point of tangency. On $(0, 4)$ the curve tends to $+\infty$ at both ends, so it has a minimum there too, at about $(1.35,\, 42.7)$. Clearing denominators in $\mathrm{f}'(x) = 0$ gives $(x+2)(x^4 - 10x^3 + 100x^2 - 72x - 64) = 0$. The quartic has exactly these two real roots, so there are no other turning points. In particular, f increases throughout $x > 4$.



The sketch is drawn to scale, which makes the dip between $x = -2$ and $x \approx -0.32$ look shallow. The lowest point there is only about 4.5 below the axis, and the curve touches the asymptote at $(-\frac{1}{2}, -\frac{9}{2})$.

Checks

- $f(-\frac{1}{2}) = -\frac{9}{2} - 0 = -\frac{9}{2}$, which is on $y = x - 4$. A numerical derivative at $x = -\frac{1}{2}$ gives 1.000.
- Expand $(x+2)^2(x^2 - 12x - 4)$ and compare with $x^2(x-4)^2 - 16(2x+1)^2$: both are $x^4 - 8x^3 - 48x^2 - 64x - 16$.
- $f(1000) - (1000 - 4) \approx -0.064 \approx -64/1000$, as the $64/x$ estimate predicts.

Common slips

- Forgetting that near $x = 0$ the curve goes to $+\infty$ on *both* sides, because of the x^2. The two vertical asymptotes behave differently.
- Proving tangency by showing only that the line meets the curve. You also need the matching gradient, or equivalently the repeated root.
- Expanding the quartic numerator in part (ii) and then trying to spot its roots, instead of seeing the difference of two squares.

Beyond STEP

A line meets a curve with a "double root" exactly when it is tangent. That is the idea of *multiplicity of intersection*, which is basic to algebraic geometry and leads to results such as Bézout's theorem on how many times two curves meet. Rational functions are approximated at infinity by their polynomial part, and the size of the remainder ($64/x$ here) tells you how fast the curve approaches the asymptote. This is a first taste of asymptotic expansions. Siklos, *Advanced Problems in Mathematics*, has more curve-sketching problems in this style.

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